TOPIC RESEARCH

Yes. I did a literature survey focused specifically on **deep-learning computer vision for Earth observation, satellite change detection, foundation models, domain shift, and environmental monitoring**.

My conclusion is that you can **keep your registered topic exactly as it is** and add a much more specific second line underneath it. More importantly, there is a credible research gap that is both **useful and realistically implementable**.

1. Keep the official topic unchanged

Your submitted topic:

> **AI-Based Computer Vision for Automated Inspection Using Deep Learning**

Add this underneath:

> **Application Focus: Automated Earth Observation and Environmental Change Analysis Using Multi-Temporal Satellite Imagery**

I would use exactly that for Phase 1.

It doesn't contradict the original topic:

Computer Vision → satellite imagery
Automated inspection → automated inspection of Earth's surface
Deep Learning → CNNs / Transformers / foundation models
Analysis → environmental change detection and quantification

You can then tell your faculty:

> *The project investigates how deep-learning-based computer vision can be used to automatically inspect multi-temporal satellite imagery and identify, quantify and interpret significant changes in Earth's surface, with applications in environmental monitoring.*

That gives you enormous freedom without changing the officially registered topic.

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2. I think your strongest research direction is this

Don't make the project simply:

> "Detect deforestation using YOLO."

That's already a well-developed area.

Instead:

> **Generalizable, label-efficient, multi-temporal environmental change detection from satellite imagery.**

In simpler terms:

Can an AI model learn what genuine environmental change looks like, rather than simply learning the visual appearance of one particular location/dataset?

This is actually a current research problem.

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3. Why ordinary satellite change detection isn't enough

Suppose you give a model:

Image A — 2020

Image B — 2025

and ask:

> "What changed?"

Sounds easy.

But the images may differ because of:

- different seasons
- different sunlight
- atmospheric conditions
- clouds/haze
- sensor differences
- different viewing angles
- vegetation cycles
- soil moisture
- image registration errors
- different geographic regions

The model may therefore report:

> **CHANGE DETECTED**

when nothing environmentally significant actually happened.

The literature calls these **pseudo-changes**.

Recent reviews explicitly identify atmospheric/radiometric variation, seasonal cycles, registration errors and sensor heterogeneity as major difficulties. (DOI)

And this is not merely an old problem.

A 2025 IEEE paper on **domain-agnostic change detection** specifically addresses the problem of models failing when the imagery comes from different domains. (DOI)

An AAAI 2026 paper reports that state-of-the-art remote-sensing change-detection models can suffer very large performance degradation—reported as **60–80%**—in unseen environments and imaging conditions. (AAAI Publications)

That is a **very good research gap for you**.

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4. The interesting research question

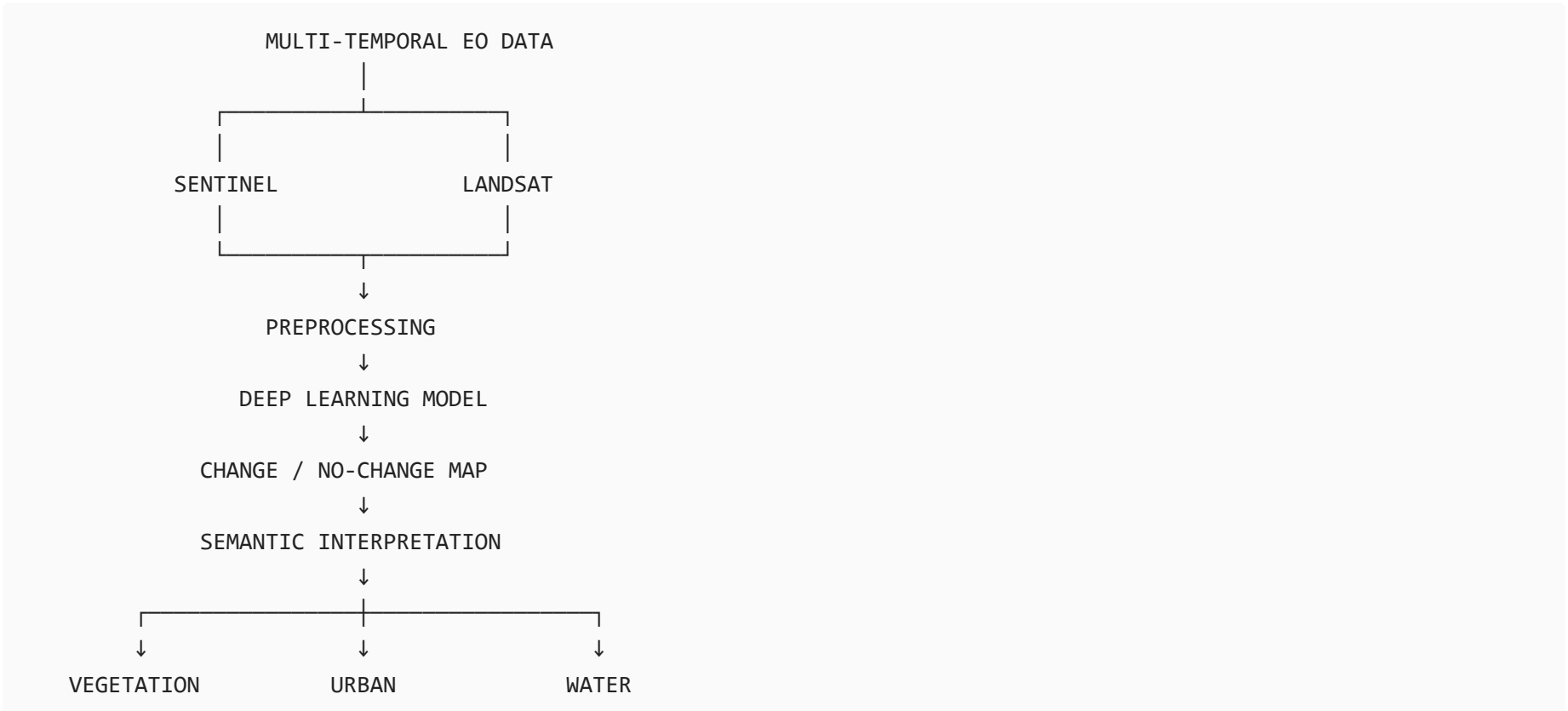
Your project could therefore ask:

> **How can deep-learning-based computer vision reliably detect genuine environmental changes in satellite imagery across different geographic regions, seasons and imaging conditions?**

Then your system isn't tied to:

- forests only
- cities only
- floods only
- agriculture only

Instead:



```
LOSS            GROWTH            CHANGE
  │               │                 │
  └───────────────┼─────────────────┘
                  ↓
          ENVIRONMENTAL REPORT
```

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5. There is a very interesting technology you can actually use

This is where the project gets considerably more modern than ordinary "CNN on satellite images."

Geospatial Foundation Models

NASA and IBM have released **Prithvi**, a geospatial foundation model specifically trained on satellite imagery.

The newer **Prithvi-EO-2.0** was trained on **4.2 million global time-series samples** from Harmonized Landsat-Sentinel-2 data and incorporates temporal and location information. It is designed to transfer to different Earth-observation tasks. (Hugging Face)

The architecture is particularly relevant to your idea because it isn't simply:

> image → CNN → classification

It uses a Vision Transformer / masked-autoencoder approach and represents **spatiotemporal satellite data**. (Hugging Face)

NASA explicitly lists applications such as:

- flood mapping
- burn-scar mapping
- crop classification
- land-use monitoring
- ecosystem monitoring
- landslide detection
- biomass estimation

for the model. (NASA Science)

So you would be working with **actual current Earth-observation AI research**, not an artificial "modern technology" bolted onto a basic project.

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6. And there is a second research gap: labels

This is another excellent area.

Traditional deep-learning change detection often requires huge quantities of labelled pixel-level data.

But creating:

```
pixel 1 = changed
pixel 2 = unchanged
pixel 3 = changed
...
```

for millions of satellite images is extremely expensive.

A 2025 review specifically devoted to **label-efficient deep-learning change detection** identifies:

- semi-supervised learning
- weak supervision
- self-supervised learning
- active learning
- few-shot learning
- unsupervised learning

as important directions because pixel-level annotation is expensive. (Wiley Online Library)

Another 2025 survey similarly concludes that practical deployment is constrained by diverse inputs and limited labelled data. (ResearchGate)

That gives you another potential research question:

> **Can we achieve useful environmental change detection with substantially less labelled satellite data by leveraging pretrained Earth-observation models and self-supervised learning?**

That's much more research-oriented than simply training YOLO.

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7. Third gap: multimodal Earth observation

This is another direction that I think is particularly promising.

You don't have to rely solely on RGB.

Satellite sensors can provide different kinds of information:

```
Optical / RGB
     +
NIR
     +
SWIR
     +
SAR
     +
Temporal information
     ↓
MULTIMODAL EARTH REPRESENTATION
```

For example, **SAR can observe through clouds and in darkness**, whereas optical imagery can provide rich spectral information. Recent reviews identify multimodal RS change detection as an important future direction. (IEEE Xplore)

That creates a useful question:

> **Can optical + SAR information reduce false environmental-change detections caused by clouds, lighting and seasonal variation?**

That's genuinely interesting.

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8. Fourth gap: don't just say "something changed"

This is where your environmental-awareness vision becomes powerful.

Traditional binary change detection:

```
CHANGE
NO CHANGE
```

isn't enough.

Imagine:

```
2012 → 2026

████████████
      ↓
   CHANGE
```

The user wants:

> **What changed?**

So you move toward **semantic change detection**:

```
Forest
   ↓
Bare land

Agriculture
```

```
   ↓
Urban

Water
   ↓
Bare land

Bare land
   ↓
Vegetation
```

Recent literature explicitly identifies **binary, multiclass and semantic change detection** as different research tasks. (IEEE Xplore)

That is much closer to your eventual OSINT vision.

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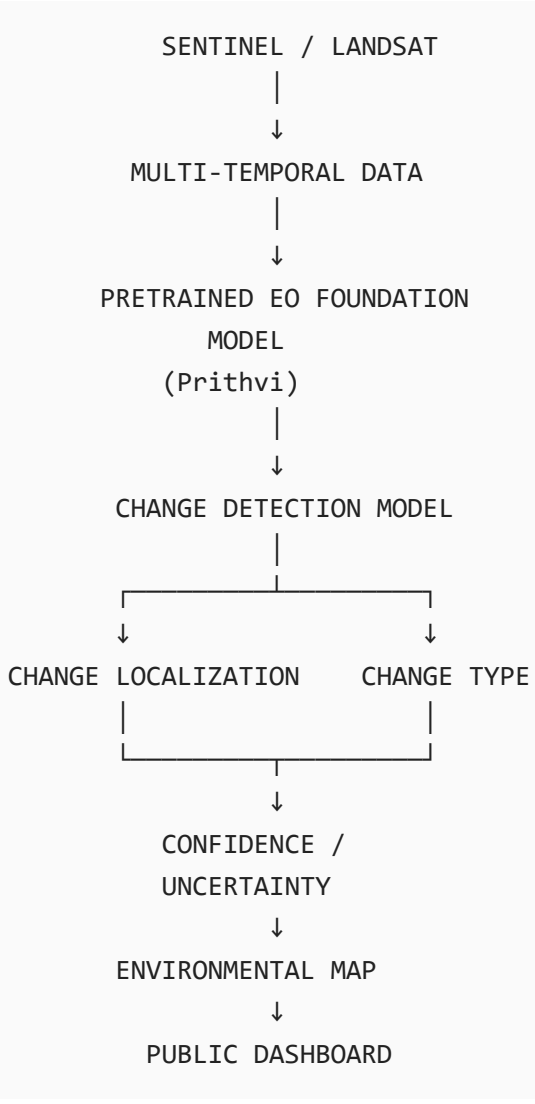
9. Put all four together

Now your potential project becomes:

Core problem

> **Reliable semantic environmental change detection from multi-temporal satellite imagery under geographic, seasonal and sensor/domain variation, with limited labelled data.**

And your potential approach:



You **do not need to implement all of this**.

That is the research space.

Your actual project can select one manageable contribution.

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10. What I'd actually recommend you build

Given that this is a college project and you need something demonstrable, I'd choose:

Domain-Generalized Semantic Change Detection

Prototype

Input:

```
Location A
2020 satellite image
+
2026 satellite image
```

Output:

```
                CHANGE MAP

Forest → Urban       4.8 km²
Forest → Bare        1.2 km²
Water → Land         0.4 km²
Urban expansion      7.3 km²

Confidence:
Forest→Urban         93%
Forest→Bare          87%
Water→Land           61%
```

Then test the model on **a different geographical region that wasn't used during training**.

That last part is crucial.

Instead of saying:

> "Our model achieved 96% accuracy on the dataset."

you can say:

> "Our model was trained on Region A and evaluated on previously unseen Region B to measure geographic generalization."

That is much more meaningful.

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11. You have plenty of genuine datasets

This isn't a project where you'll spend three months collecting data.

Sentinel-2 provides global multispectral imagery with a designed revisit frequency of about five days at the equator, and its applications include land monitoring, agriculture, forestry, climate change and disaster management. (Copernicus Data Space Ecosystem)

There are established research datasets as well.

For example, **S2Looking** contains 5,000 bitemporal image pairs and more than 65,920 annotated change instances, deliberately introducing difficult viewing-angle and illumination variation. (MDPI)

You also have established datasets such as LEVIR-CD and other change-detection benchmarks covered extensively by recent surveys.

So your concern about finding data is basically solved.

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12. The literature progression is actually excellent for your Phase 1

You can structure your first 10 papers approximately like this:

Area	What you learn
Traditional remote-sensing change detection	Problem foundations
CNN/Siamese CD	Baseline deep learning
BIT	Transformers for temporal/spatial context
ChangeFormer	Transformer-based Siamese architecture
S2Looking	Difficult real-world imaging conditions
Domain adaptation	Handling distribution shifts
Domain generalization	Generalizing to unseen regions
Label-efficient CD	Reducing annotation requirements

Area	What you learn
Multimodal CD	Optical + SAR/etc.
Prithvi-EO	Modern geospatial foundation models

For example, **BIT** introduced a bitemporal Transformer specifically to model spatial-temporal context efficiently. (arXiv)

ChangeFormer subsequently used a Transformer-based Siamese architecture for multi-scale long-range change detection. (arXiv)

Then the field moved toward:

Transformers → domain generalization → label efficiency → multimodal learning → geospatial foundation models.

That's an excellent literature-survey storyline.

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13. One thing I would NOT claim

Don't make your academic project:

> **"AI that predicts Earth's future catastrophes."**

That's too broad and scientifically difficult.

Satellite CV can **observe and quantify indicators**, but predicting something like:

> "This region will become environmentally catastrophic in 15 years"

requires climate models, causal relationships, socioeconomic variables, weather data, hydrology, etc.

Instead:

What your CV system can legitimately do

Observe

> Forest decreased 14%.

Detect

> New built-up region appeared.

Compare

> Vegetation changed significantly between 2010 and 2026.

Quantify

> 23 km² of land-cover transition detected.

Track

> Change trajectory over multiple years.

Flag

> Region exhibits abnormal environmental change.

Then, potentially later:

Risk estimation

> Region shows indicators associated with increased flood/fire/deforestation risk.

That is much more scientifically defensible.

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14. And this connects directly to your huge OSINT idea

Your college prototype could be:

> **AI-based environmental change detector**

But the eventual platform could become:

> **Earth OSINT — an open environmental intelligence system**

with:

```
                EARTH OSINT
                    │
    ┌───────────────┼───────────────┐
    ↓               ↓               ↓
  FOREST          URBAN          WATER
    ↓               ↓               ↓
DEFOREST.        SPRAWL         FLOOD
    │               │               │
    └───────────────┼───────────────┘
                    ↓
           MULTI-TEMPORAL AI
                    ↓
           CHANGE DETECTION
                    ↓
           ENVIRONMENTAL
              INDICATORS
                    ↓
           HUMAN-READABLE
              EVIDENCE
```

That is where your original vision of **showing people what is happening to Earth** fits.

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15. My recommendation for the actual project framing

Official topic

> **AI-Based Computer Vision for Automated Inspection Using Deep Learning**

Subtopic

> **Application Focus: Automated Earth Observation and Environmental Change Analysis Using Multi-Temporal Satellite Imagery**

Problem statement

> **To develop a deep-learning-based computer vision system capable of detecting, quantifying and semantically interpreting environmental changes from multi-temporal satellite imagery, with emphasis on robustness to geographic, seasonal and imaging variations.**

Research direction

> **Domain-generalized and label-efficient semantic change detection using modern Earth-observation foundation models.**

Long-term vision

> **An open environmental OSINT platform that converts satellite observations into accessible evidence of planetary change.**

The important conclusion from the literature survey

Yes, there is a real research problem here.

But I would **not** try to invent a completely new CV architecture.

The smarter contribution for your team is:

Existing CV capability: satellite change detection
↓
Existing technology: Transformers / foundation models
↓
Known weakness: geographic/domain shift + seasonal pseudo-change + expensive labels

↓
Your contribution: improve robustness/generalization and semantic interpretation
↓
Application: environmental monitoring
↓
Interface: public Earth-OSINT visualization

That gives you a project that is simultaneously **CV research + deep learning + satellite data + modern foundation models + environmental usefulness**, while remaining compatible with the topic you already submitted.

And the literature is current enough that you can genuinely discuss **2025–2026 research**, rather than building a project around an outdated CNN-only approach. Recent reviews specifically identify **label-efficient learning, multimodal change detection, lightweight models, foundation-model-assisted change detection, and vision-language approaches** as active directions. (DOI)

One particularly strong candidate for us to investigate next is:

> **“Can Prithvi-EO-2.0 + a lightweight change-detection head generalize to environmental changes in geographic regions it was never trained on?”**

That is concrete enough to become an actual experiment, rather than just a big website idea.